Parth Natekar

IIT Madras | University of California, San Diego

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Education

INSTITUTION & DEGREE	CGPA	YEAR
University of California, San Diego *MS, Electrical and Computer Engineering* *(Focus Areas: Machine Learning, Medical Devices)*	**3.95/4.0**	Expected March '23
Indian Institute of Technology, Madras *Bachelor of Technology in Engineering Design,* *Master of Technology in Biomedical Design*	**8.89/10.00**	2020
Fergusson College *Higher Secondary School (Major: Electronics)*	92.4 %	2015

Publications

Yiding Jiang, **Parth Natekar**, et al. "Methods and Analysis of The First Competition in Predicting Generalization of Deep Learning." *Proceedings of the NeurIPS 2020 Competition and Demonstration Track, PMLR 133:170-190* (Link).

Parth Natekar, Manik Sharma. "Representation Based Complexity Measured for Predicting Generalization in Deep Learning." *Winning Solution of the NeurIPS 2020 competition on Predicting Generalization in Deep Learning, arXiv:2012.02775 (2020)* (Link).

Zichen Wang, **Parth Natekar**, et al. "MitoTNT: Mitochondrial Temporal Network Tracking for 4D live-cell fluorescence microscopy data." *Under review in PLOS Comp. Biology* (Preprint).

Avinash Kori, **Parth Natekar**, Balaji Srinivasan, and Ganapathy Krishnamurthi. "Abstracting Deep Neural Networks into Concept Graphs for Concept Level Interpretability." ***Best Paper Award**, AAAI 2021 Health Intelligence Workshop* (Link).

Parth Natekar, Avinash Kori, and Ganapathy Krishnamurthi. "Demystifying Brain Tumour Segmentation Networks: Interpretability and Uncertainty Analysis." *Frontiers in Computational Neuroscience 14 (2020): 6* (Link).

Research & Professional Experience

Intuitive Surgical, Sunnyvale

Computer Vision/AI Intern | DeepSight Team, Future Forward Research

(Manager: Omid Mohareri, Manager, Applied Research, CV/AI) — *June '22 - September '22*

- Built, from scratch, a **deep-learning-based, kinematics informed, 3D hand-tracking and pose estimation pipeline** for supporting tele-operated surgery on the Da Vinci surgical robot.
- Ideated on and developed the entire pipeline, from sensor setup and data collection, to deep learning model training, to 3D prediction, to multi-camera sensor fusion and kinematic correction.
- Demonstrated working use cases of the pipeline during surgical activities on the Da Vinci robot.

University of California, San Diego

Researcher | Advanced Robotics and Controls Lab, UCSD

(Guide: Dr. Michael Yip, UC San Diego) — *June '21 - Present*

- Built a **deep-learning-based 3D human-pose estimation framework using RGB-D data** for dynamic 6-degree-of-freedom annotation tracking in mixed reality telementoring scenarios.
- Lead developer and researcher for **Mercury, a mixed-reality based platform which enables remote, mobile, tele-operated robotic surgery** (built in Python, ROS, C#, and Unity).
- Conceptualized and built the entire pipeline for Mercury, including **teleoperation kinematics, VR headset to ROS to robot networking, and the VR interface** for the robotic simulator.

Research Associate | 4DCELL Lab, UCSD

(Guide: Dr. Johannes Schoeneberg, UC San Diego) — *September '21 - Present*

- Working on **deep learning for 4D (3D + time) microscopic imaging** and **algorithm development for cell-organelle tracking** to better understand neuro-degenerative diseases.

Anheuser-Busch InBev, Bangalore

Data Scientist | Machine Learning for Compliance Analytics *Aug '20 - May '21*

- Worked on building, prototyping, and testing machine learning models and developing hypotheses to flag anomalous and risky transactions for legal and financial compliance.

Medical Imaging and Reconstruction Lab, IIT Madras

Research Associate | Interpretable Deep Learning for Medical Image Analysis

(Guide: Dr. Ganapathy Krishnamurthi, Engineering Design, IIT Madras) June '19 - June '20

- Built a pipeline for **interpretability and uncertainty estimation of deep learning models** which perform various image processing tasks in the medical domain, such as tumor segmentation.
- This involved developing a new method for concept-level interpretability as well as implementing techniques like network dissection and bayesian dropout on segmentation and classification models.
- Implemented the pipeline on benchmark brain tumor segmentation datasets to understand information flow and representation in internal layers and to quantify uncertainty of model predictions.
- Developed a python package for direct implementation of the proposed interpretability pipeline on deep neural networks for any relevant biomedical dataset.

Honeywell Technology Solutions, Bangalore

Research Intern | Graphical Modelling of Neural Networks for Causal Inference

(Mentor: Satyanarayan Kar, Honeywell Aerospace, India) *January '19 - June '19*

- Devised a probabilistic graphical model based algorithm for **causal inference on deep neural networks, to provide human-understandable explanations** of what the network learns.
- Implemented the algorithm on benchmark datasets and successfully delineated variables in the model which were causal for a particular prediction compared to those with spurious correlations.
- Developed a deep learning pipeline for time-series analysis of flight data in order to predict anomalous flight landings. Achieved 86% accuracy in predicting instability 2 minutes prior to landing.
- Implemented an algorithm for object detection and text extraction from aircraft documents.

Convergence Design Lab, Purdue University, USA

Research Intern | Wearables for Human-Computer Interaction

(Guide: Dr. Karthik Ramani, Mechanical Engineering, Purdue University) *May '18-July '18*

- Completed the entire design cycle – ideation, detail design, fabrication and prototyping, of IOT based flexible, wearable electronic devices for interaction with Mixed Reality environments.
- Devised a novel method to design flexible electronic circuits on various fabric substrates. Conducted material and sensor tests to integrate tactile sensing using carbon based piesoresistive sensors.
- Developed multiple ergonomic, working prototypes. Set up networking capabilities via UDP/TCP protocols and tested the wearables in Mixed Reality and IoT environments.
- Created applications in Unity and C# to validate the efficiency of using the prototyped wearables.
- Reduced task time by 44% and error rate by 74% vis-à-vis traditional interaction methods.

CERN (European Organization for Nuclear Research), Geneva

Engineering Intern | Testing Apparatus for CMS cooling system

(Mentor: Antti Onnela, EP-DT, CERN, Geneva) *May '17 - July '17*

- Selected to be part of a 3-member team from IIT Madras to work on the Large Hadron Collider.
- Designed an adiabatic testing apparatus for the evaporative CO_2 based cooling system of the CMS Detector using concepts from thermal, mechanical and electrical design.
- Created a software interface in LabView to enable easy user interaction.

Awards & Achievements

- **Winner, NeurIPS competition on Predicting Generalization in Deep Learning**, 2020
- **PURE fellow**: Awarded PURE Scholarship for a two-month Research Internship at Purdue University, Indiana, having been selected in the top 14 among 800 students from IIT Madras.
- **CBSE Class X School Topper**: Came first in school with percentage of 97.00%. Awards for highest marks in Math (100/100) and Science (99/100).
- One of 15 students selected from 5000 by Infosys for their **'Spark Program**.

Skills

Programming: Python, Deep Learning API (Keras, Tensorflow, Pytorch), ROS, C, C++, C#, OpenCV, JavaScript, Unity, Arduino, Raspberry Pi, LaTeX, SQL
Languages: Competent French, Limited Working German.

Related Course Work

Computation & Machine Learning

- Advanced Computer Vision
- Statistical Learning
- Unsupervised Learning
- Machine Learning for Engineering Applications
- Probabilistic Graphical Modelling
- Human-Centered AI
- Computational Evolutionary Biology

Mechanical/Electrical Engineering

- Design of Mechanical Systems (I & II)
- Design of Electronic Systems (I & II)
- Digital Signal Processing
- Engineering Mechanics
- Design of Thermal and Fluid Systems
- Electromagnetic Compatibility for Design
- Bio-Micro-Electrical-Mechanical Systems

Biomedical Design

- Medical Image Analysis
- Design of Diagnostic and Monitoring Systems
- Design of Implantable and Surgical Devices
- Medical Equipment Dissection Lab
- Human Anatomy, Physiology & Biomechanics
- Biomaterials Engineering

Design Methodology

- Functional and Conceptual Design
- Human Factors in Design
- Geometric Modelling and CAD
- Computational Methods in Design
- Design for Form and Aesthetics
- Creative Design

Relevant Course Projects

Computational Modelling of Pathogen Interactomes

January to May 2018 | Supervisor : Dr. Karthik Raman

- Delineated drug-resistance pathways in the pathogen Staph Aureus through systems level computational modelling of the protein-protein interaction network of the organism.
- Devised a modified network-based shortest path algorithm by integrating node centrality measures with chemical protein interaction scores between drug molecules and proteins.
- Achieved 68.5% prediction accuracy on already known resistance pathways, validating the viability of our algorithm for exploring drug resistance pathways in pathogens.

Ultrasound Nerve Image Segmentation

Jan to May 2017 | Supervisor : Dr. Ganapathy Krishnamurthi

- A deep learning-based project for semantic segmentation of ultrasound nerve cell images.
- Used a fully-convolutional neural-network implemented in Python using Keras and Tensorflow.
- Obtained 71.2% Dice Score on test dataset – on par with then state-of-the-art techniques

Teaching and Voluntary Experience

Teaching Assistant - Design of Electronic Systems Lab, IIT Madras

Professor :Dr. Tuhin Subhra Santra — *Aug '19 - Dec '19*

- Selected as Graduate Teaching Assistant for the Design of Electronic Systems Lab (Fall 2019).
- Managed and guided a group of 60+ students in electronic circuit design principles.

Teaching Assistant - Medical Image Analysis, IIT Madras

Professor :Dr. Ganapathy Krishnamurthi — *Aug '19 - Dec '19*

- Conducted tutorial sessions to teach students the programming aspects of Medical Image Analysis.

Volunteering

- Worked with an organization called *Lata Mahajan Foundation* on their project *EducationLinked*, which is aimed at overall development of tribal and underprivileged children.
- Worked with *iVolunteer*, Bangalore to help organize health camps in orphanages.